Feature Mutual Representation-Based Graph Domain Adaptive Network for Unsupervised Hyperspectral Change Detection

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***Abstract*— Recently, deep neural networks (DNNs) have been widely used in hyperspectral image change detection (HSI-CD). Generally, training such a DNN-based HSI-CD network often requires a large number of labeled training samples. However, it is time-consuming, labor-intensive, or even infeasible to label training samples in practice. In this article, we propose a feature mutual representation-based graph domain adaptive network (FGDANet) for unsupervised HSI-CD. This method constructs a pseudosiamese backbone consisting of two customized unsupervised learning domains, which can make full use of the information from different domains through the graph domain adaptation strategy to improve the feature expression capability and generalization. There are three key characteristics: first, in each customized unsupervised learning domain, a graph convolutional network (GCN)-based difference feature extraction architecture is designed to model the local and global dependence among the features of multitemporal HSIs; second, a progressive graph-to-pixel joint constraint strategy (PJCS) is proposed to provide the high-confidence training sample labels for the unsupervised learning of the network in each domain; and third, the homogeneous mutual representation joint graph feature alignment (HJGFA) module of the graph domain adaptation strategy can make full use of the difference features from the two domains through the information interaction to facilitate the model to capture the changed and unchanged essential characteristics. The experimental results on four HSI datasets demonstrate the superiority of the proposed FGDANet. Code is available at https://github.com/Jiahuiqu/FGDANet.**

***Index Terms*— Domain adaptive, graph convolutional network (GCN), hyperspectral image change detection (HSI-CD), unsupervised.**

I. Introduction

With the development of remote sensing imaging technology, lots of remote sensing images can be acquired. Compared with other remote sensing images, hyperspectral images (HSIs) with wider spectral coverage and higher spectral resolution can reflect the property of the ground object accurately [1], [2]. Due to the aforementioned advantages, HSIs have been the important data source for earth observation in the field of the remote sensing image processing [3], [4], [5], [6], [7]. HSI change detection (HSI-CD), a hot topic in the field of remote sensing image processing, is a technique that can obtain the changes of the earth surface by analyzing the characteristics of multitemporal HSIs covering the same area. It has shown great application value and potential in many practical fields, such as natural disaster detection [8], urbanization monitoring [9], and environmental monitoring [10].

So far, various HSI-CD methods have been proposed, among which the traditional methods can be mainly divided into three categories, namely image algebra-based methods, image transformation-based methods, and image classification-based methods. The image algebra-based methods that are developed in early days directly process the image at pixel-level and detect changes through image difference, image ratio, or image regression analysis. The typical image algebra-based method, named change vector analysis (CVA) [11], adopts the image difference operation to obtain the difference map of the multitemporal image and delineates a threshold to judge whether the pixels belong to changed or unchanged class. The image transformation-based methods extract representative features from an original image which can effectively reflect the characteristics of the image and highlight the distinguishability of the targets to improve the accuracy of CD. Principal component analysis (PCA) [12] and iteratively reweighted multivariate alteration detection (IR-MAD) [13] belong to the image transformation-based methods. The image classification-based methods often regard the CD process as a classification task. Support vector machines (SVMs) [14], tree-based clustering [15], and extreme learning machines

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(ELMs) [16] have been used in CD tasks. These traditional CD methods have played an important role in promoting the development of remote sensing CD techniques. However, traditional methods cannot fully utilize the information of images due to the limited feature extraction capability, which lead to poor performance.

Due to the strong capabilities in feature extraction, deep learning has been widely applied in the field of remote sensing image processing [17], [18], [19], [20]. Many deep learning-based CD methods have achieved excellent performance [21]. Mou et al. [22] proposed a CD method called ReCNN, in which convolutional neural networks (CNNs) are used to extract spatial–spectral features, and recurrent neural networks (RNNs) are used to model the temporal dependences of multitemporal multispectral images. Zhan et al. [23] adopted 1-D convolution and 2-D convolution to extract spectral and spatial features of multitemporal HSIs, respectively, and to determine changed and unchanged classes. Yang et al. [24] employed spatial–spectral residual attention to capture multilevel and multiscale features of HSIs and to improve the detection ability of the network for complex regions. Wang et al. [25] proposed a triplet transformer framework (TriTF) for HSI-CD, which can make full use of the spatial–spectral information and temporal information of the images through a "branch–backbone–branch" structure to improve the detection accuracy. Li et al. [26] proposed a cross-band 2-D self-attention network (CBANet), which fully considered the dependence between adjacent pixels and enhanced the spatial–spectral features through the 2-D self-attention module to achieve efficient HSI-CD. Luo et al. [27] proposed a multiscale diff-changed feature fusion network for HSI-CD, which used the bidirectional diff-changed feature representation and multiscale attention fusion to generate the discriminative spatial–spectral–temporal features for detection performance improvement.

Although deep learning-based methods have been proven effective with good automatic feature learning capabilities, they often require large amounts of labeled samples for network training. However, it is a very time-consuming and labor-intensive task to label datasets manually in practical application. For HSIs, manual sample labeling task is more difficult or even infeasible due to the wide coverage area and harsh environment, which greatly limits the number of available training samples. In addition, it is difficult for CNN-based CD methods that perform convolution operations with fixed and rectangular convolutional kernels to fully exploit local and global spatial–spectral information of HSIs with complex and irregular objects.

To address the above problems, in this article, we propose a feature mutual representation-based graph domain adaptive network (FGDANet) for unsupervised HSI-CD. As an unsupervised task, to obtain high-confidence training samples, the node-level and pixel-level spatial–spectral correlations of multitemporal HSIs are jointly considered on the graph structure with irregular topology to improve the reliability of the training samples. Meanwhile, the GCNs with large-scale receptive fields instead of CNNs with only local receptive fields are employed to extract difference features of multitemporal HSIs. Through the graph node aggregation operation, GCN can not only capture the local and global information of the image but also effectively maintain the class boundary of the target regions in the image, thereby promoting the identification of targets in HSIs. To further improve the generalization and detection accuracy of the network, inspired by the idea of domain adaptation, a pseudosiamese unsupervised backbone is adopted to realize the interaction guidance of difference features for different unsupervised learning domains to enhance the feature expression ability of FGDANet. In summary, the main contributions of this article are given as follows.

1) A graph domain adaptive HSI-CD network is proposed to achieve accurate unsupervised learning for different domains, which presents a pseudosiamese unsupervised backbone to achieve cross-domain interaction of local and global difference features of multitemporal HSIs to enhance the feature representation ability of the network and improve the detection accuracy.
2) A customized unsupervised learning framework is proposed, in which a progressive graph-to-pixel joint constraint strategy (PJSC) is proposed to screen out reliable samples with high confidence on graph structure for the unsupervised learning of the network.
3) A homogeneous mutual representation joint graph feature alignment (HJGFA) module is proposed to exploit difference features from two different unsupervised learning domains simultaneously to improve generalization of the FGDANet. Among them, the homogeneous graph structures are presented to realize mutual linear representation of same-class graph nodes in different domains to promote cross-domain information interaction and improve network performance.

The remainder of this article is organized as follows. Section II describes the related work; Section III analyzes the proposed FGDANet in detail; Section IV discusses the experiments; and finally, Section V concludes this article.

II. Related Work

CNNs have been widely used in many image processing tasks, which utilize fixed-size convolution kernels to integrate features in regular regions around the central pixel. However, the limited receptive field makes it difficult to extract global information, which leads to insufficient utilization of features [28], [29]. To address the aforementioned limitation, GCNs that incorporate graph structures into convolution operations have been successfully proposed. GCNs aggregate the information between the central graph node and its adjacent nodes by performing convolution operations in non-Euclidean space. By aggregating the graph node features at different distances, the local and global information of the image can be captured, and the feature expression ability of the convolution operation can be improved. In recent years, many studies on GCNs have been successfully proposed [30], [31], [32]. Jia et al. [33] presented a graph-in-graph (GiG) model for hyperspectral classification, in which internal and external graph convolutions are used to extract local and global features

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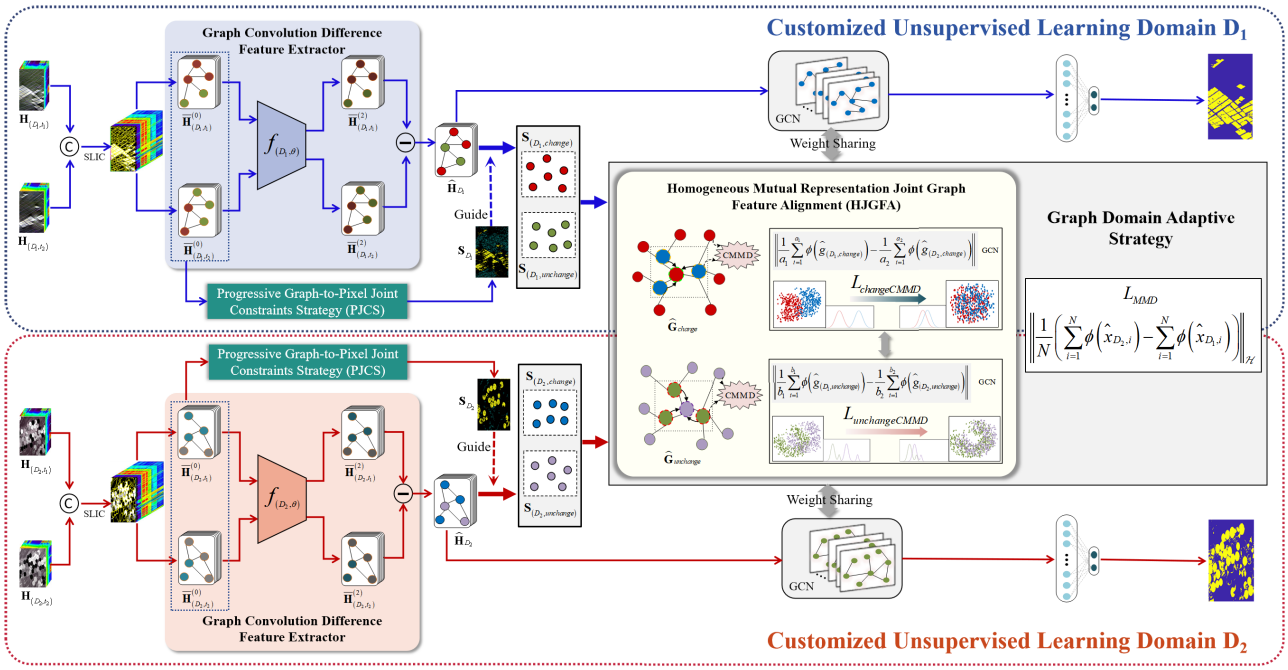


Fig. 1. Overview of the proposed FGDANet.

of HSI to improve the classification accuracy. Gao et al. [34] proposed a dynamic graph representation learning method based on GCN to fully extract the global structure information of dynamic graphs and improve the accuracy of hyperspectral classification.

In GCN, an undirected graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$ usually consists of graph node set $\mathbf{V} = \{v_1, v_2, v_3, \ldots, v_N\}$ and edge set $\mathbf{E}$, where N is the number of graph nodes and $v_i (i \in \{1, 2, 3, \ldots, N\})$ represents a graph node. The adjacency matrix $\mathbf{A}$ is defined to reflect the connection relationship between graph nodes. For the input signal $\mathbf{x}$ corresponding to each graph node v, a filter g_θ is used to perform the convolution operation in the Fourier domain as

$$g_\theta * \mathbf{x} = \mathbf{U} g_\theta \mathbf{U}^{\mathrm{T}} \mathbf{x} \tag{1}$$

where $\mathbf{U}$ represents the matrix composed of the eigenvectors of the normalized Laplacian matrix $\mathbf{L}_{\mathrm{sym}} = \mathbf{U}\Lambda\mathbf{U}^{\mathrm{T}}$. Here, Λ is the diagonal matrix composed of the eigenvalues of $\mathbf{L}_{\mathrm{sym}}$, and g_θ expressed as $g_\theta(\Lambda)$ can be regarded as a function of Λ. Although the graph convolution operation is successfully implemented in (1), the high computational complexity brought by the matrix decomposition of $\mathbf{L}_{\mathrm{sym}}$ is unacceptable. Therefore, Hammond et al. [35] adopted K-order Chebyshev polynomials to parameterize the convolution kernel g_θ to accelerate the graph convolution operation as

$$g_{\theta'}(\Lambda) \approx \sum_{k=0}^{K} \theta'_k T_k\left(\widetilde{\Lambda}\right) \tag{2}$$

where k is the order of Chebyshev polynomial, $\widetilde{\Lambda} = 2/(\lambda_{\max})\Lambda - \mathbf{I}$ with θ' being the vector of Chebyshev coefficients and $\lambda_{\max}$ being the largest eigenvalue of Laplacian matrix $\mathbf{L}$, $T_k(\cdot)$ is the Chebyshev polynomial with $T_k(\mathbf{x}) = 2\mathbf{x}T_{k-1}(\mathbf{x}) - T_{k-2}(\mathbf{x})$, $T_0(\mathbf{x}) = 1$, and $T_1(\mathbf{x}) = \mathbf{x}$. It is worth noting that the K-order Chebyshev polynomials represent feature aggregation between a central graph node and its K-order adjacency graph nodes. In general, to simplify the calculation, only the first-order neighbors ($K = 1$) are considered, and (1) can be simplified as

$$g_{\theta'} * \mathbf{x} \approx \theta'_0\mathbf{x} + \theta'_1(\mathbf{L} - \mathbf{I})\mathbf{x} = \theta'_0\mathbf{x} - \theta'_1\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\mathbf{x} \tag{3}$$

where θ'_0 and θ'_1 are free parameters, $\mathbf{I}$ is an identity matrix, and $\mathbf{D}$ is the degree matrix with $\mathbf{D}_{i,i} = \sum_j \mathbf{A}_{i,j}$. To further reduce the amount of parameters and avoid overfitting, let $\theta = \theta'_0 = -\theta'_1$, then (3) can be rewritten as

$$g_\theta * \mathbf{x} = \theta\left(\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)\mathbf{x}. \tag{4}$$

Since the value range of $(\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)})$ is $[0, 2]$, iteratively performing graph convolution operations as the network deepens will lead to numerical instability or even gradient explosion. To address this problem, Kipf and Welling [36] performed a renormalization trick $\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)} \rightarrow \widetilde{\mathbf{D}}^{-(1/2)}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-(1/2)}$ with $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\widetilde{\mathbf{D}} = \sum_j \widetilde{\mathbf{A}}_{ij}$. In this way, the graph convolution process can be expressed as

$$\mathbf{H}^{(l)} = \sigma\left(\widetilde{\mathbf{A}}\mathbf{H}^{(l-1)}\mathbf{W}^{(l)}\right) \tag{5}$$

where $\mathbf{H}^{(l)}$ represents the output of the lth graph convolutional layer, $\sigma(\cdot)$ is the activation function, and $\mathbf{W}^{(l)}$ represents the learnable weight matrix of the lth graph convolutional layer. Based on the graph structure, GCN can realize the aggregation of graph node features and extract the features in a non-Euclidean space.

III. Proposed Method

A. Overall Framework

As shown in Fig. 1, the proposed FGDANet employs an unsupervised pseudosiamese structure as the backbone,

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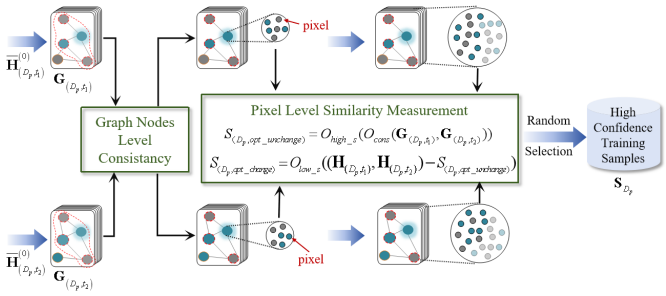


Fig. 2. Illustration of the PJSC for high-confidence training sample selection.

which consists of two customized unsupervised learning domains. Specifically, in each customized unsupervised learning domain, a graph convolution difference feature extractor is designed to generate local and global difference features that are crucial for CD, and a PJCS is proposed to provide highly reliable training samples for the unsupervised learning of networks. Furthermore, graph domain adaptive strategy realizes the interaction of difference features between different domains under the guidance of high-confidence training samples generated by PJCS, which can effectively enhance the feature expression ability and generalization of the proposed FGDANet.

B. Customized Unsupervised Learning Framework

1) Graph Convolution Difference Feature Extractor: In general, the most typical implementation of CD is to use the difference features extracted from multitemporal HSIs to train a classifier and determine whether the target at the corresponding location has changed. The extraction of difference features of HSIs is the most fundamental and critical step for CD. Considering that the target regions in HSIs are diverse and irregular in shape, we design a graph convolution difference feature extractor to explore the local-global features and boundary information of target regions, and then obtain discriminative difference information through difference operation.

In each customized unsupervised learning domain D_p, where $p \in \{1, 2\}$ represents the domain number, the input HSI is denoted as $\mathbf{H}_{(D_p,t_q)}$, where $q \in \{1, 2\}$ is the phase number. In order to perform graph convolution to extract local and global features of images, multitemporal HSIs $\mathbf{H}_{(D_p,t_1)}$ and $\mathbf{H}_{(D_p,t_2)}$ are transformed into graph structures. For the graph construction process, the most critical step is the determination of the graph nodes $\mathbf{V}_{(D_p,t_q)}$ and the edges $\mathbf{E}_{(D_p,t_p)}$. In this article, to construct the graph structures for multitemporal HSIs with the same node distribution, $\mathbf{H}_{(D_p,t_1)}$ and $\mathbf{H}_{(D_p,t_2)}$ are cascaded along the channel dimension. Then, the cascaded image is segmented by using simple linear iterative clustering (SLIC) [37] algorithm where local k-means clustering is used to divide the cascaded image into a series of superpixel regions as

$$\mathbf{X}_{D_p} = \mathbf{M}_{D_p}^{\mathrm{T}} \cdot \left[\mathbf{H}_{(D_p,t_1)} \parallel \mathbf{H}_{(D_p,t_2)}\right] \tag{6}$$

where $\mathbf{X}_{D_p}$ is the result of superpixel segmentation of the cascaded image for domain D_p, $\mathbf{M}_{D_p}$ is an association matrix that reflects the mapping relationship between superpixel regions and the original image, and $\parallel$ is the cascade operation. Then, the obtained $\mathbf{X}_{D_p}$ is split along the channel dimension to obtain the superpixel segmentation results of HSIs $\mathbf{H}_{(D_p,t_1)}$ and $\mathbf{H}_{(D_p,t_2)}$. In each HSI, each superpixel region is regarded as a graph node $v^i_{(D_p,t_q)}$, and the value of each graph node is the average of the pixel values in the superpixel region. A series of homogeneous superpixel regions constitutes the graph node set $\mathbf{V}_{(D_p,t_q)}$. Then, the edge set $\mathbf{E}_{(D_p,t_p)}$ that can reflect the connection relationship between the graph nodes is usually represented by an adjacency matrix $\mathbf{A}_{(D_p,t_q)}$. The adjacency matrix $\mathbf{A}_{(D_p,t_q)}$ is determined by calculating the similarity between graph nodes as

$$\mathbf{A}^{i,j}_{(D_p,t_q)} = \begin{cases} e^{-\frac{\left\|v^i_{(D_p,t_q)} - v^j_{(D_p,t_q)}\right\|^2}{\sigma^2}} \\ \qquad v^i_{(D_p,t_q)} \in Nei\left(v^j_{(D_p,t_q)}\right) \\ \qquad \text{or } v^j_{(D_p,t_q)} \in Nei\left(v^i_{(D_p,t_q)}\right) \\ 0, \quad \text{otherwise} \end{cases} \tag{7}$$

where σ is a hyperparameter set to 0.2 according to experience, and $\mathrm{Nei}(v^i_{(D_p,t_q)})$ and $\mathrm{Nei}(v^j_{(D_p,t_q)})$ represent the sets of neighbors of node $v^i_{(D_p,t_q)}$ and $v^j_{(D_p,t_q)}$, respectively. In this article, the top m ($m = 30$) graph nodes in the whole graph that are most similar to the central graph node are selected as adjacent nodes. The elements in the adjacency matrix $\mathbf{A}_{(D_p,t_q)}$ effectively represent the weights of edges in the graph. In this way, the graph structure $\mathbf{G}_{(D_p,t_q)} = (\mathbf{V}_{(D_p,t_q)}, \mathbf{E}_{(D_p,t_q)})$ is successfully constructed by considering the spatial–spectral relationship of the HSIs.

The graph convolution feature extractor can extract non-Euclidean features by aggregating information from graph nodes and adjacent nodes, which facilitates the discrimination of irregular target regions in HSI. The feature extraction process can be formulated as follows:

$$\begin{aligned} \overline{\mathbf{H}}^{(L)}_{(D_p,t_q)} &= f_{(D_p,\theta)}\left(\mathbf{G}_{(D_p,t_q)}\right) \\ &= \sigma\left(\widetilde{\mathbf{A}}_{(D_p,t_q)} \cdots \sigma\left(\widetilde{\mathbf{A}}_{(D_p,t_q)} \cdots \sigma\left(\widetilde{\mathbf{A}}_{(D_p,t_q)}\right.\right.\right. \\ &\quad \left.\left.\left. \times \mathbf{G}_{(D_p,t_q)}\mathbf{W}^{(0)}_{(D_p,t_q)}\right) \cdots \mathbf{W}^{(l)}_{(D_p,t_q)}\right) \cdots \mathbf{W}^{(L-1)}_{(D_p,t_q)}\right) \end{aligned} \tag{8}$$

where $l \in (0, L)$, $\overline{\mathbf{H}}^{(L)}_{(D_p,t_q)}$ is the feature of the HSI at time t_q in the pth domain extracted from the Lth GCN layer, $\sigma(\cdot)$ is the linear rectifier unit (ReLU), $f_{(D_p,\theta)}(\cdot)$ represents the graph convolution feature extractor with the parameter θ in the pth domain, $\mathbf{G}_{(D_p,t_q)}$ is the graph structure of the HSI at time t_q in the pth domain, $\widetilde{\mathbf{A}}_{(D_p,t_q)} = \mathbf{A}_{(D_p,t_q)} + \mathbf{I}$ with $\mathbf{A}_{(D_p,t_q)}$ as the adjacency matrix of the HSI at time t_q in the pth domain, and $\mathbf{W}^{(l)}_{(D_p,t_q)}$ denotes the learnable matrix of the lth GCN layer for the HSI at time t_q in the pth domain. In this way, the graph convolutional network (GCN) can perform convolution operations in irregular image regions by constructing flexible graph topologies, which can capture the features of target regions with various geometric distributions in HSI. As the network deepened, the graph convolution realizes the transmission of local and global information between graph nodes

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and enhances image feature representations while maintaining the shape properties of the target regions.

The purpose of CD is to distinguish differences in multitemporal HSIs. Therefore, in the graph convolution difference feature extractor, the features extracted by GCN are differentiated to obtain the difference features with significant distinguishability as follows:

$$\widehat{\mathbf{H}}_{D_p} = \overline{\mathbf{H}}^{(L)}_{(D_p,t_1)} - \overline{\mathbf{H}}^{(L)}_{(D_p,t_2)} \tag{9}$$

where $\widehat{\mathbf{H}}_{D_p}$ is the difference feature of the HSIs in the pth domain. Sufficient difference features can provide a powerful data source for detecting changes to improve the detection accuracy.

2) PJSC for High-Confidence Training Sample Selection: The performance of deep learning-based models heavily relies on the quantity and quality of training samples. However, it is worth noting that manual annotation for inaccessible target regions in practical applications presents significant challenges. Even if only a small proportion of the area can be labeled, it is still very time-consuming and labor-intensive. To address this problem, as shown in Fig. 2, we propose a PJCS to select high-confidence training samples for network training, which progressively considers the consistency between graph nodes in graph topology and the similarity between pixels within each graph node. On the graph node level, we first select the highly similar graph nodes with consistent first-order adjacent nodes between the two images. Then, on the pixel level, the highly similar pixels are selected as the optional unchanged training samples from the selected node pairs, and the unsimilar pixels between the two images are selected as the optional changed training samples. In addition, to make the model learn more features, we select samples randomly from the optional changed and unchanged training samples as the final training samples. The proposed strategy can ensure the selected training samples with high confidence.

Specifically, we consider the correlation of image pixels from the graph node-level and pixel-level successively to select the changed and unchanged training samples. On the node level, the highly similar graph nodes with consistent connections of the first-order adjacent nodes are initially selected between the two images. Furthermore, on the pixel level, the Euclidean distance between the corresponding pixels inside the obtained highly similar graph nodes is calculated to further select the highly similar pixels as the optional unchanged training samples $\mathbf{S}_{(D_p,\text{opt_unchange})}$. After removing the optional unchanged training samples from the pixel set of the original images, the similarity between the remaining pixels is quantified using the Euclidean distance, and the pixels with low similarity are selected as the optional changed training samples $\mathbf{S}_{(D_p,\text{opt_change})}$. This process can be described as follows:

$$\begin{cases} \mathbf{S}_{(D_p,\text{opt_unchange})} = O_{\text{high_}s}\left(O_{\text{cons}}\left(\mathbf{G}_{(D_p,t_1)}, \mathbf{G}_{(D_p,t_2)}\right)\right) \\ \mathbf{S}_{(D_p,\text{opt_change})} = O_{\text{low_}s}\left(\left(\mathbf{S}_{(D_p,t_1)}, \mathbf{S}_{(D_p,t_2)}\right) - \mathbf{S}_{(D_p,\text{opt_unchange})}\right) \end{cases} \tag{10}$$

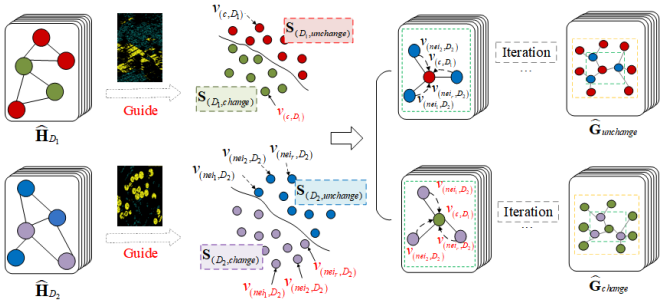


Fig. 3. Construction process of homogeneous graph structures.

where $\mathbf{G}_{(D_p,t_1)}$ and $\mathbf{G}_{(D_p,t_2)}$ are the graphs corresponding to multitemporal HSIs $\mathbf{H}_{(D_p,t_1)}$ and $\mathbf{H}_{(D_p,t_2)}$, $O_{\text{cons}}(\cdot)$ represents the graph node connection consistency constraint operation at the node level, $O_{\text{high_}s}(\cdot)$ denotes the pixel-level Euclidean similarity measure operation to select high similarity pixels, $\mathbf{S}_{(D_p,t_1)}$ and $\mathbf{S}_{(D_p,t_2)}$ are the pixel sets corresponding to multitemporal HSIs $\mathbf{H}_{(D_p,t_1)}$ and $\mathbf{H}_{(D_p,t_2)}$, and $O_{\text{low_}s}(\cdot)$ is the pixel-level Euclidean similarity measure operation to select low similarity pixels. It can be found that, in PJSC, the distribution consistency of graph nodes and the Euclidean distance between pixels in the graph nodes are comprehensively considered to select the training samples. In this way, we can obtain the optional training samples with high confidence.

In addition, to make the model learn more features, a certain proportion of samples are randomly selected from the optional changed and unchanged training samples as the final training samples $\mathbf{S}_{D_p}$

$$\mathbf{S}_{D_p} = \text{Rand}\left(\mathbf{S}_{(D_p,\text{opt_change})}, \mathbf{S}_{(D_p,\text{opt_unchange})}\right) \tag{11}$$

where $\text{Rand}(\cdot)$ represents the random selection operation. In this way, the high-confidence training samples $\mathbf{S}_{D_p}$ of the unsupervised learning domains D_p are selected for optimization of network parameters.

C. Graph Domain Adaptation Strategy

Most existing methods perform the model training with the samples from a single dataset, which may result in the model being tailored to the specific characteristics of that dataset and lead to poor generalization. Consequently, it is beneficial to incorporate the samples from diverse datasets for model training to make the model learn the general features. However, due to the differences in illumination, atmosphere, and other external conditions, spectral shift exists between the same ground objects of different datasets, which leads to the distribution discrepancy between different datasets. The model trained with the samples that exist in distribution discrepancy may learn the inconsistent features of the same object and generate the inaccurate detection results. In this article, we propose a graph domain adaptation strategy to reduce the feature distribution discrepancy between the datasets of different domains, which can make the model fully utilize the information from different datasets and learn the changed and unchanged essential characteristics across domains. This approach can improve both feature expression capability and generalization of the CD model.

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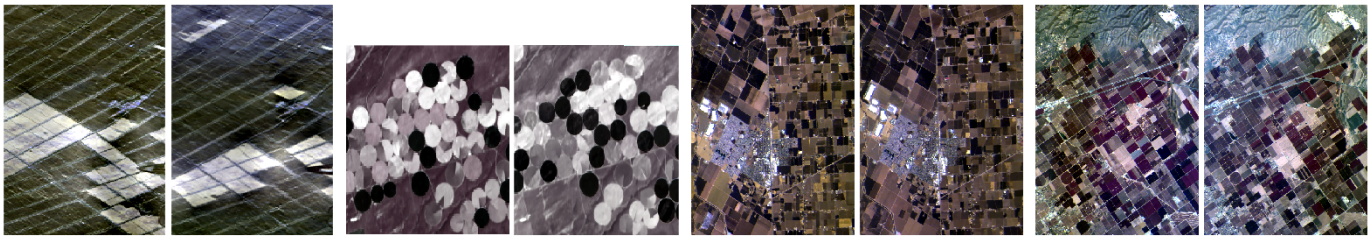


Fig. 4. Illustration of datasets. (a1) Image of China dataset acquired on May 3, 2006. (a2) Image of China dataset acquired on April 23, 2007. (b1) Image of USA dataset acquired on May 1, 2004. (b2) Image of USA dataset acquired on May 8, 2007. (c1) Image of Bay Area dataset acquired in 2013. (c2) Image of Bay Area dataset acquired in 2015. (d1) Image of Santa Barbara dataset acquired in 2013. (d2) Image of Santa Barbara dataset acquired in 2014.

1) Homogeneous Mutual Representation Joint Graph Feature Alignment: The HJGFA module is proposed to make full use of the local and global difference features from the two domains through the information interaction, which can facilitate the model to capture changed and unchanged essential characteristics across domains. Specifically, taking the difference features obtained by graph convolution difference feature extractors in two domains as input, the HJGFA module can align the feature distributions of the two domains, which allows the network to exploit sufficient difference features of the two domains simultaneously. To aggregate the difference features from the two domains to reduce the distribution shift, the changed and unchanged homogenous graph structures are constructed, respectively, which are guided by the high-confidence training sample labels generated by PJCS. As shown in Fig. 3, the graph nodes of the difference feature in each domain are divided into two sets based on the labels obtained through PJCS, i.e., the set of changed nodes $\mathbf{S}_{(D_p,\text{change})}$ and the set of unchanged nodes $\mathbf{S}_{(D_p,\text{unchange})}$. Since the construction process of the changed and unchanged homogenous graph structures is the same, we only illustrate the graph structure establishment of the changed homogenous graph structure in detail. Specifically, guided by the labels of high-confidence training samples from two different domains, the changed node $v_{(c,D_p)}$ in one domain is chosen as the central node and the changed graph nodes $\{v_{(\text{nei}_1,D_{\bar{p}})}, v_{(\text{nei}_2,D_{\bar{p}})}, \cdots, v_{(\text{nei}_r,D_{\bar{p}})}\}$ with $r = 15$ is randomly chosen from the other domain as its first-order adjacent nodes, where $(p, \bar{p}) = \{(1, 2), (2, 1)\}$. Continuing to iteratively treat each adjacent node as a new central node, the changed homogeneous graph structure $\widehat{\mathbf{G}}_{\text{change}}$ is established in a domain until all of them are completely exhausted. Following the same steps, the unchanged homogeneous graph structure $\widehat{\mathbf{G}}_{\text{unchange}}$ of unchanged nodes is established. On the established graph structure, the aggregation operation of GCN can effectively realize the mutual linear representation of the same class of graph nodes in two domains as follows:

$$v_{(c,D_p)} = \sum_{i=1}^{r} \omega_i v_{(\text{nei}_i,D_{\bar{p}})} + o_i \tag{12}$$

where ω_i is the weight coefficient, and o_i represents the bias. Through the interaction of the node information in different domains, the obtained GCN weight parameters are applicable to both domain D_1 and domain D_2.

In addition, the conditional maximum mean discrepancy (CMMD) is proposed to effectively narrow the difference between homogeneous graph node features of two domains. Specifically, $L_{\text{changeCMMD}}$ is used to measure the distance between the changed node $\widehat{g}_{(D_1,\text{change})_i}$ from domain D_1 and the changed node $\widehat{g}_{(D_2,\text{change})_i}$ from domain D_2 of the changed graph structure $\widehat{\mathbf{G}}_{\text{change}}$ in reproducing kernel Hilbert space (RKHS). $L_{\text{unchangeCMMD}}$ is used to measure the distance between the unchanged node $\widehat{g}_{(D_1,\text{unchange})_i}$ from domain D_1 and the unchanged node $\widehat{g}_{(D_2,\text{unchange})_i}$ from domain D_2 of the unchanged graph structure $\widehat{\mathbf{G}}_{\text{unchange}}$ in RKHS. Finally, the total CMMD constraint between graph nodes in these two domains can be computed as follows:

$$\begin{aligned} &L_{\text{CMMD}} \\ &= L_{\text{changeCMMD}} + L_{\text{unchangeCMMD}} \\ &= \left\| \frac{1}{a_1}\sum_{i=1}^{a_1} \phi\left(\widehat{g}_{(D_1,\text{change})_i}\right) - \frac{1}{a_2}\sum_{i=1}^{a_2} \phi\left(\widehat{g}_{(D_2,\text{change})_i}\right) \right\|_{\mathcal{H}} \\ &\quad + \left\| \frac{1}{b_1}\sum_{i=1}^{b_1} \phi\left(\widehat{g}_{(D_1,\text{unchange})_i}\right) - \frac{1}{b_2}\sum_{i=1}^{b_2} \phi\left(\widehat{g}_{(D_2,\text{unchange})_i}\right) \right\|_{\mathcal{H}} \end{aligned} \tag{13}$$

where $\phi(\cdot)$ is the mapping from the original feature space to the RKHS, $\|\cdot\|_{\mathcal{H}}$ is RKHS norm, a_1 and a_2 are the numbers of changed samples in domain D_1 and domain D_2 of $\widehat{\mathbf{G}}_{\text{change}}$, respectively, and b_1 and b_2 are the numbers of unchanged samples in domain D_1 and domain D_2 of $\widehat{\mathbf{G}}_{\text{unchange}}$, respectively. Conditional distribution adaptation of difference features between two domains can be achieved by minimizing L_{CMMD}. Through the weight-sharing mechanism, the obtained GCN parameters are directly applied to the graph convolution operation on the difference features in two domains. The graph convolution operation with shared weights is performed on the difference features $\widehat{\mathbf{H}}_{D_1}$ and $\widehat{\mathbf{H}}_{D_2}$ to obtain the deep features $\widehat{\mathbf{X}}_{D_1}$ and $\widehat{\mathbf{X}}_{D_2}$.

2) Marginal Distribution Alignment: To further minimize the distance of the data distributions between the two domains, not only the conditional distributions of the graph nodes belonging to the same class from the two domains but also the marginal distributions of $\widehat{\mathbf{X}}_{D_1}$ and $\widehat{\mathbf{X}}_{D_2}$ are aligned. The marginal distribution alignment function L_{MMD} can be shown

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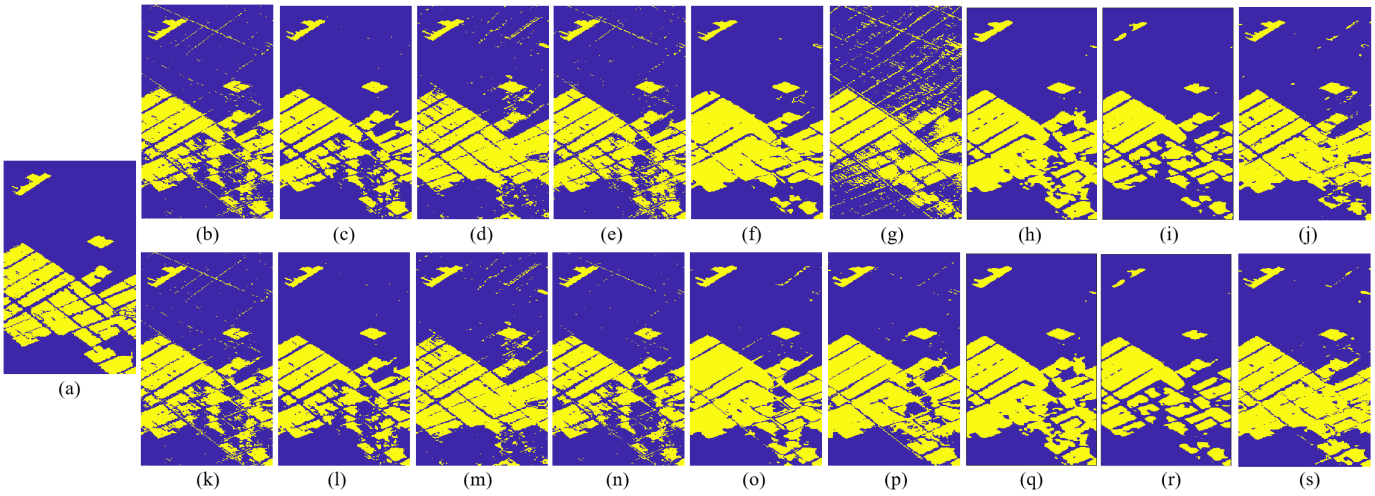


Fig. 5. CD results of different methods on China dataset under different numbers of training samples. (a) Ground-truth map. (b) CVA (0.5%). (c) PCA (0.5%). (d) IR-MAD (0.5%). (e) SVM (0.5%). (f) BCNNs (0.5%). (g) ReCNN (0.5%). (h) SGCL (0.5%). (i) HyperNet (0.5%). (j) FGDANet (0.5%). (k) CVA (20%). (l) PCA (20%). (m) IR-MAD (20%). (n) SVM (20%). (o) BCNNs (20%). (p) ReCNN (20%). (q) SGCL (20%). (r) HyperNet (20%). (s) FGDANet (20%).

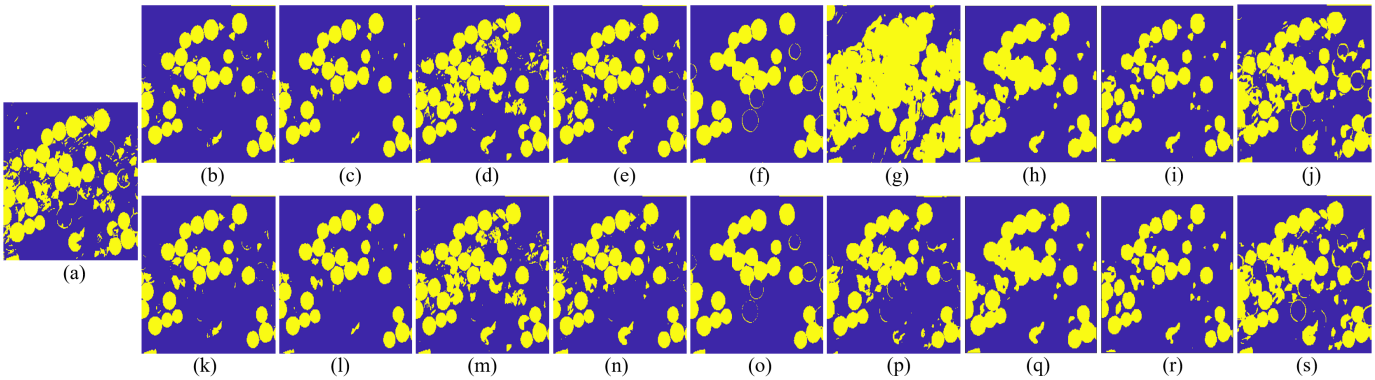


Fig. 6. CD results of different methods on USA dataset under different numbers of training samples. (a) Ground-truth map. (b) CVA (0.5%). (c) PCA (0.5%). (d) IR-MAD (0.5%). (e) SVM (0.5%). (f) BCNNs (0.5%). (g) ReCNN (0.5%). (h) SGCL (0.5%). (i) HyperNet (0.5%). (j) FGDANet (0.5%). (k) CVA (20%). (l) PCA (20%). (m) IR-MAD (20%). (n) SVM (20%). (o) BCNNs (20%). (p) ReCNN (20%). (q) SGCL (20%). (r) HyperNet (20%). (s) FGDANet (20%).

as

$$L_{\mathrm{MMD}} = \left\| \frac{1}{n}\sum_{i=1}^{n}\phi\left(\widehat{x}_{D_2,i}\right) - \frac{1}{n}\sum_{i=1}^{n}\phi\left(\widehat{x}_{D_1,i}\right) \right\|_{\mathcal{H}} = \left\| \frac{1}{n}\left(\sum_{i=1}^{n}\phi\left(\widehat{x}_{D_2,i}\right) - \sum_{i=1}^{n}\phi\left(\widehat{x}_{D_1,i}\right)\right) \right\|_{\mathcal{H}} \tag{14}$$

where n is the number of high-confidence training samples and $\widehat{x}_{D_p,i}$ is the graph node in feature map $\widehat{\mathbf{X}}_{D_p}$.

Finally, by jointly considering the graph node conditional distribution adaptation and the feature edge distribution adaptation, the HJGFA can improve the generalization of the proposed FGDANet with the idea of domain adaptation.

D. Network Training

In order to discriminate the changes of the multitemporal HSIs, the difference features of domains D_1 and D_2 after feature alignment are, respectively, sent to the classifier composed of the fully connected layer to obtain the final CD results $\mathbf{R}_{D_1}$ and $\mathbf{R}_{D_2}$. To improve the discriminative ability of the network, the widely used cross-entropy loss L_{ce} is used to measure the gap between the detection result $\mathbf{R}_{D_p}$ and the high-confidence training sample labels $\mathbf{S}_{D_p}$ as

$$L_{\mathrm{ce},D_p} = -\frac{1}{n}\sum_{j=1}^{n}\left(\mathbf{R}_{(D_p,j)}\log\mathbf{S}_{(D_p,j)} + \left(1-\mathbf{R}_{(D_p,j)}\right)\log\left(1-\mathbf{S}_{(D_p,j)}\right)\right) \tag{15}$$

where $\mathbf{R}_{(D_p,j)}$ and $\mathbf{S}_{(D_p,j)}$ represent the label values of the selected high-confidence training samples and predicted values of detection map, respectively. Using the high-confidence labels obtained by PJCS, the loss function L_{FGDANet} jointly composed of conditional feature alignment loss L_{CMMD}, marginal distribution feature alignment loss L_{MMD}, and cross-entropy loss L_{ce} is adopted to train the proposed FGDANet, which can be expressed as

$$L_{\mathrm{FGDANet}} = \alpha L_{\mathrm{CMMD}} + \beta L_{\mathrm{MMD}} + \lambda\left(L_{\mathrm{ce},D_1} + L_{\mathrm{ce},D_2}\right) \tag{16}$$

where the weight coefficients α, β, and λ are e^{-1}, e^{-1}, and 1, respectively. During the network training process, the loss

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TABLE I
QUANTITATIVE ANALYSIS OF DIFFERENT METHODS FOR CHINA AND USA DATASETS

Training sample(%)	Index	CVA	PCA	IR-MAD	SVM	BCNNs	ReCNN	SGCL	HyperNet	FGDANet
China										
0.5	**OA**	0.8749	0.8892	0.8974	0.8742	0.9097	0.8037	0.8950	0.9023	**0.9128**
	KC	0.6998	0.7322	0.7632	0.6912	0.7923	0.5522	0.7626	0.7617	**0.7996**
1	**OA**	0.8749	0.8892	0.8974	0.8761	0.8841	0.8457	0.8950	0.9023	**0.9216**
	KC	0.6998	0.7322	0.7632	0.6993	0.7433	0.6681	0.7626	0.7617	**0.8228**
5	**OA**	0.8749	0.8892	0.8974	0.8797	0.8916	0.9020	0.8950	0.9023	**0.9241**
	KC	0.6998	0.7322	0.7632	0.7076	0.7565	0.7849	0.7626	0.7617	**0.8296**
10	**OA**	0.8749	0.8892	0.8974	0.8774	0.9025	0.9019	0.8950	0.9023	**0.9261**
	KC	0.6998	0.7322	0.7632	0.7019	0.7812	0.7848	0.7626	0.7617	**0.8321**
20	**OA**	0.8749	0.8892	0.8974	0.8740	0.9041	0.8926	0.8950	0.9023	**0.9258**
	KC	0.6998	0.7322	0.7632	0.6921	0.7837	0.7572	0.7626	0.7617	**0.8330**
USA										
0.5	**OA**	0.8789	0.8723	0.9111	0.8972	0.8547	0.6872	0.8908	0.8870	**0.9166**
	KC	0.7024	0.6831	0.7968	0.7515	0.6447	0.4200	0.7452	0.7237	**0.8093**
1	**OA**	0.8789	0.8723	0.9111	0.8955	0.8568	0.7831	0.8908	0.8870	**0.9171**
	KC	0.7024	0.6831	0.7968	0.7467	0.6495	0.5338	0.7452	0.7237	**0.8056**
5	**OA**	0.8789	0.8723	0.9111	0.9046	0.8578	0.9061	0.8908	0.8870	**0.9219**
	KC	0.7024	0.6831	0.7968	0.7709	0.6490	0.7848	0.7452	0.7237	**0.8223**
10	**OA**	0.8789	0.8723	0.9111	0.8935	0.8497	0.8783	0.8908	0.8870	**0.9210**
	KC	0.7024	0.6831	0.7968	0.7414	0.6359	0.7104	0.7452	0.7237	**0.8178**
20	**OA**	0.8789	0.8723	0.9111	0.8976	0.8581	0.8860	0.8908	0.8870	**0.9239**
	KC	0.7024	0.6831	0.7968	0.7524	0.6512	0.7278	0.7452	0.7237	**0.8291**

L_{FGDANet} is minimized by updating the network parameters until the detection performance of the network is optimal.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description

The first HSI dataset used in the experiments is China, which covers an area of farmland in Yancheng, Jiangsu, China. As shown in Fig. 4(a1) and (a2), the multitemporal HSIs contained in the China dataset were captured by Earth Observing-1 (EO-1) Hyperion sensor on May 3, 2006 and April 23, 2007, respectively. Each HSI with 450 × 140 pixels covers 155 bands in the spectral range of 0.4–2.5 μm.

The second dataset named USA was captured over the irrigated farmland area in Hermiston, Umatilla, Oregon, USA, on May 1, 2004, and May 8, 2007, respectively. The pseudo-color images of multitemporal HSIs contained in this dataset are shown in Fig. 4(b1) and (b2), with a size of 200 × 240 pixels. Each image acquired with the EO-1 Hyperion sensor covered 114 bands ranging from 0.4 to 2.5 μm.

The third dataset named Bay Area, shown in Fig. 4(c1) and (c2), was captured by Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 2013 and 2015, respectively. The HSIs contained in this dataset show the distribution of buildings and crops over San Francisco Bay Area in the city of Patterson, California, USA. Each HSI contains 224 spectral bands with a size of 600 × 500 pixels.

As shown in Fig. 4(d1) and (d2), the fourth dataset named Santa Barbara was captured by the AVIRIS sensor over the Santa Barbara region in California, USA, in which two HSIs reflect the land coverage of this area in 2013 and 2014, respectively. The spatial size of each HSI in this dataset is 984 × 740 pixels with 224 spectral bands.

B. Evaluation Criteria and Implementation Details

To verify the effectiveness of the proposed method, we select eight CD methods as the comparison methods, e.g., CVA [11], PCA [12], IR-MAD [13], SVM [14], ReCNN [22], BCNNs [38], SGCL [39], and HyperNet [40]. These algorithms cover various categories of the existing state-of-the-art CD methods. CVA, PCA, IR-MAD, and SVM belong to the traditional methods. ReCNN and BCNNs are two supervised deep learning-based methods. SGCL and HyperNet are two state-of-the-art unsupervised deep learning-based methods.

The evaluation metrics overall accuracy (OA) and kappa coefficient (KC) for pixel-level analysis are used to measure the CD performance. OA, with a value ranging from 0 to 1, is used to measure the proportion of correctly classified pixels to the total number of pixels. KC is introduced to describe the consistency between the predicted map and ground truth map. The higher the value of KC within the range of [0,1], the more effective CD will be. In this article, the experiments are performed on NVIDIA GTX 3090 GPU with PyTorch framework. The Adam optimizer with a learning rate of 5.0×10^{-5} is adopted in experiments. The number of training epochs is 700. In FGDANet, the whole image is sent into the network without the need of batch size. To construct the initial graph structures, the superpixel segmentation scales corresponding to the China, USA, Bay Area, and Santa Barbara datasets are 10, 20, 100, and 150, respectively. In graph convolution difference feature extractor, we use two-layer GCNs to extract the local and global features of HSI. For the homogeneous graph structures $\widehat{\mathbf{G}}_{\text{change}}$ and $\widehat{\mathbf{G}}_{\text{unchange}}$, the number of neighbor nodes around the central graph node is set to 15.

C. Experimental Result and Analysis

To evaluate the effectiveness of the proposed method, we conduct comparative experiments with several existing

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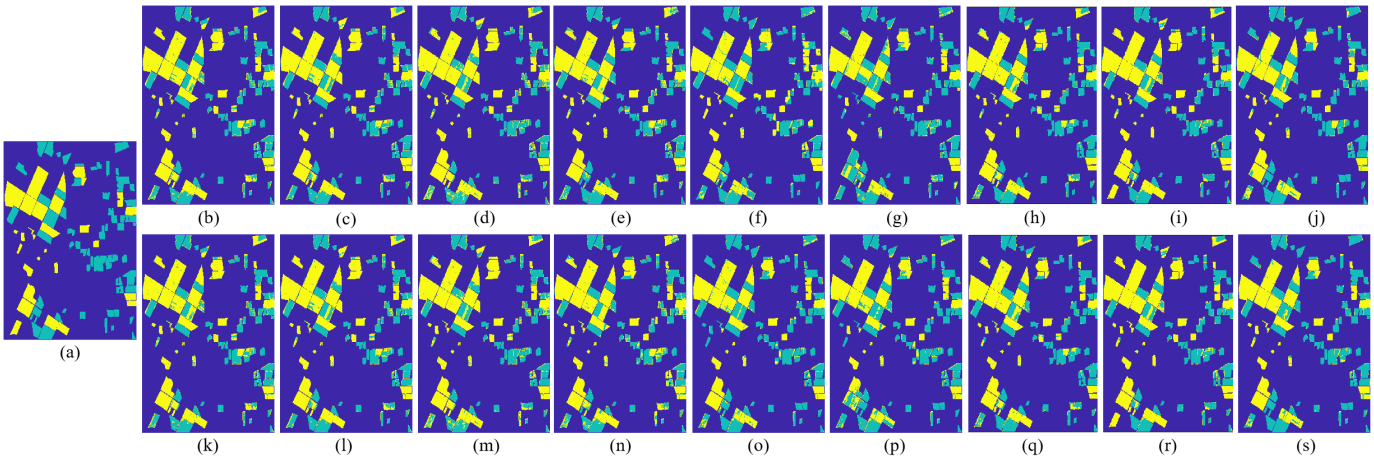


Fig. 7. CD results of different methods on Bay Area dataset under different numbers of training samples. (a) Ground-truth map. (b) CVA (0.5%). (c) PCA (0.5%). (d) IR-MAD (0.5%). (e) SVM (0.5%). (f) BCNNs (0.5%). (g) ReCNN (0.5%). (h) SGCL (0.5%). (i) HyperNet (0.5%). (j) FGDANet (0.5%). (k) CVA (20%). (l) PCA (20%). (m) IR-MAD (20%). (n) SVM (20%). (o) BCNNs (20%). (p) ReCNN (20%). (q) SGCL (20%). (r) HyperNet (20%). (s) FGDANet (20%).

TABLE II

QUANTITATIVE ANALYSIS OF DIFFERENT METHODS FOR BAY AREA AND SANTA BARBARA DATASETS

Training sample(%)	Index	CVA	PCA	IR-MAD	SVM	BCNNs	ReCNN	SGCL	HyperNet	FGDANet
Bay Area										
0.5	**OA**	0.8407	0.8422	0.8371	0.8280	0.8355	0.8355	0.8829	0.8896	**0.8995**
	KC	0.6845	0.6876	0.6782	0.6606	0.6587	0.6685	0.7670	0.7814	**0.7824**
1	**OA**	0.8407	0.8422	0.8371	0.8220	0.7793	0.7996	0.8829	0.8896	**0.9000**
	KC	0.6845	0.6876	0.6782	0.6490	0.5649	0.5952	0.7670	0.7814	**0.7845**
5	**OA**	0.8407	0.8422	0.8371	0.8283	0.8749	0.8659	0.8829	0.8896	**0.9022**
	KC	0.6845	0.6876	0.6782	0.6609	0.7948	0.7309	0.7670	0.7814	**0.7899**
10	**OA**	0.8407	0.8422	0.8371	0.8388	0.8521	0.8475	0.8829	0.8896	**0.9055**
	KC	0.6845	0.6876	0.6782	0.6815	0.6978	0.6923	0.7670	0.7814	**0.7985**
20	**OA**	0.8407	0.8422	0.8371	0.8341	0.8696	0.8753	0.8829	0.8896	**0.9088**
	KC	0.6845	0.6876	0.6782	0.6723	0.7354	0.7494	0.7670	0.7814	**0.7921**
Santa Barbara										
0.5	**OA**	0.8325	0.7933	0.8630	0.8874	0.8915	0.8806	0.8896	0.8918	**0.8966**
	KC	0.6519	0.5449	0.7099	0.7642	0.7755	0.7524	0.7763	0.7653	**0.7906**
1	**OA**	0.8325	0.7933	0.8630	0.8913	0.8824	0.8828	0.8896	0.8918	**0.8995**
	KC	0.6519	0.5449	0.7099	0.7722	0.7785	0.7589	0.7763	0.7653	**0.7966**
5	**OA**	0.8325	0.7933	0.8630	0.8921	0.8915	0.8961	0.8896	0.8918	**0.9023**
	KC	0.6519	0.5449	0.7099	0.7742	0.7743	0.7804	0.7763	0.7653	**0.8012**
10	**OA**	0.8325	0.7933	0.8630	0.8947	0.8846	0.8866	0.8896	0.8918	**0.9051**
	KC	0.6519	0.5449	0.7099	0.7833	0.7679	0.7618	0.7763	0.7653	**0.8056**
20	**OA**	0.8325	0.7933	0.8630	0.8969	0.8674	0.8849	0.8896	0.8918	**0.9171**
	KC	0.6519	0.5449	0.7099	0.7836	0.7342	0.7580	0.7763	0.7653	**0.8304**

methods on four datasets divided into two groups. Experiments are performed with different numbers of training samples (i.e., 0.5%, 1%, 5%, 10%, and 20%) generated by PJCS, and the CD results at ratios of 0.5% and 20% are visualized.

1) Experiment on China and USA Datasets: Given the China and USA datasets as inputs simultaneously, the visual results are shown in Figs. 5 and 6. For China dataset, it can be seen from Fig. 5 that there is a serious missed detection in the CD maps obtained by CVA, PCA, and SVM methods with different numbers of training samples. The subjective visual CD results generated by the four competing deep learning-based methods are also unacceptable, as they incorrectly detect many unchanged pixels as changed. For the USA dataset, as shown in Fig. 6, the traditional methods cannot detect a large number of changed pixels, and ReCNN is very sensitive to the number of training samples. It is noteworthy that the proposed FGDANet has superior feature extraction and discrimination capabilities compared to other methods, and the obtained CD maps are the closest to the ground-truth maps for both China and USA datasets. Table I shows the detection accuracy of all methods on China and USA datasets with different numbers of training samples. It can be seen that the proposed FGDANet can achieve most excellent results with different numbers of training samples. Since the shape of the change areas in China and USA datasets is relatively regular, the proposed method can achieve the optimized performance when the training samples reach a certain proportion. When the proportion of samples continues to increase, the detection results may fluctuate around the optimal result. This illustrates that a small number of training samples is enough for the model to learn the change rule.

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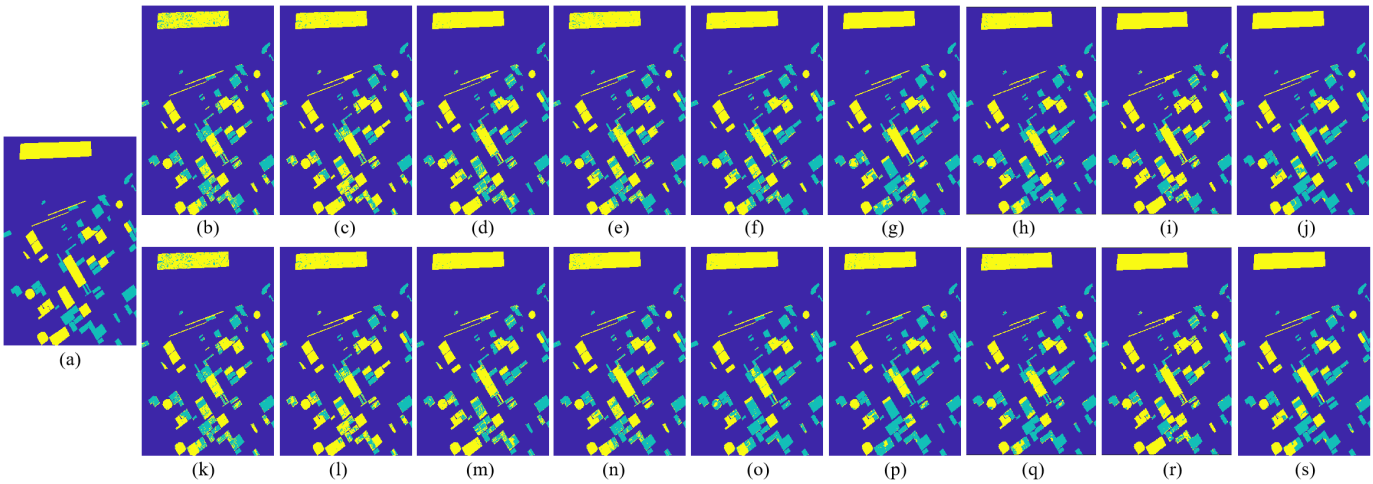


Fig. 8. CD results of different methods on Santa Barbara dataset under different numbers of training samples. (a) Ground-truth map. (b) CVA (0.5%). (c) PCA (0.5%). (d) IR-MAD (0.5%). (e) SVM (0.5%). (f) BCNNs (0.5%). (g) ReCNN (0.5%). (h) SGCL (0.5%). (i) HyperNet (0.5%). (j) FGDANet (0.5%). (k) CVA (20%). (l) PCA (20%). (m) IR-MAD (20%). (n) SVM (20%). (o) BCNNs (20%). (p) ReCNN (20%). (q) SGCL (20%). (r) HyperNet (20%). (s) FGDANet (20%).

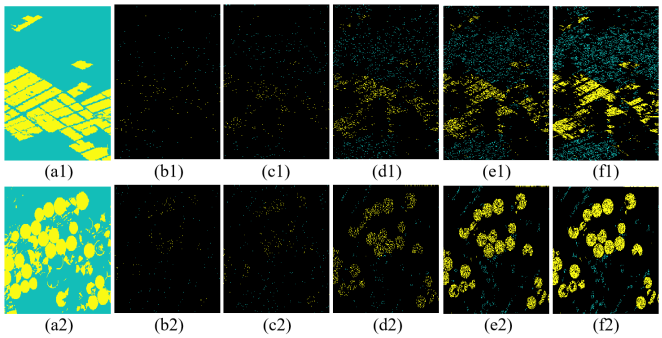


Fig. 9. Visual results of different numbers of samples obtained by the PJSC for high-confidence training sample selection on China and USA datasets. (a1) and (a2) Ground-truth map. (b1) and (b2) 0.5%. (c1) and (c2) 1%. (d1) and (d2) 5%. (e1) and (e2) 10%. (f1) and (f2) 20%.

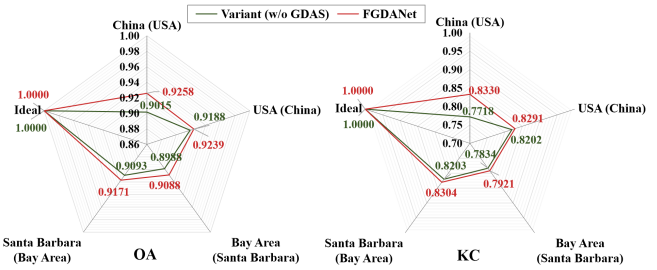


Fig. 10. CD results of FGDANet and its variant with single domain on the four datasets. China (USA) represents the output of the model that corresponds to the China dataset, and the model is trained with the samples from the China and USA datasets. The same principle applies to USA (China), Bay Area (Santa Barbara), and Santa Barbara (Bay Area).

2) Experiment on Bay Area and Santa Barbara Datasets: As Bay Area and Santa Barbara datasets are simultaneously input into the FGDANet as a group, we discuss their results together, which are presented in Figs. 7 and 8. For some subtle scattered regions, the traditional methods have poor detection performance on the two datasets. In the CD maps generated by ReCNN and BCNNs, a large number of unchanged pixels are determined to be changed. SGCL and HyperNet have relatively serious missed detection. Compared with other methods, the proposed FGDANet with the best CD performance has the least missed detection and false detection. To comprehensively verify the effectiveness of the proposed method, the quantitative experimental results on the Bay Area and Santa Barbara datasets are listed in Table II. Obviously, with different numbers of training samples, FGDANet can achieve the highest detection accuracy in terms of OA and KC, demonstrating the effectiveness and stability of the proposed FGDANet.

In summary, compared with other competing methods, the proposed method demonstrates the highest OA and KC on four widely used datasets in terms of quantitative evaluation. The visualized results illustrate that the proposed method can achieve a low rate of missing detection and false detection. In addition, it is noteworthy that the best results can also be obtained in the case of limited samples. The reason is that the proposed method can fully utilize the local and global difference features from different domains to promote the learning of change rules, so that it can improve the feature expression ability of the CD model. Additionally, the proposed pseudosiamese unsupervised backbone is constructed by GCN, where the pixels within each graph node are considered to belong to the same category. The label of the graph node can be inferred as long as the label of one pixel is available. Therefore, the proposed method reduces the reliance on a large number of training samples and still achieves excellent performance even with limited samples. Above all, the proposed method can achieve better detection performance compared with other competing methods.

D. Ablation Studies

1) Effectiveness of the PJSC for High-Confidence Training Sample Selection: In order to verify the effectiveness of the PJCS for high-confidence training sample selection, we visualize different numbers of samples obtained by PJCS on the

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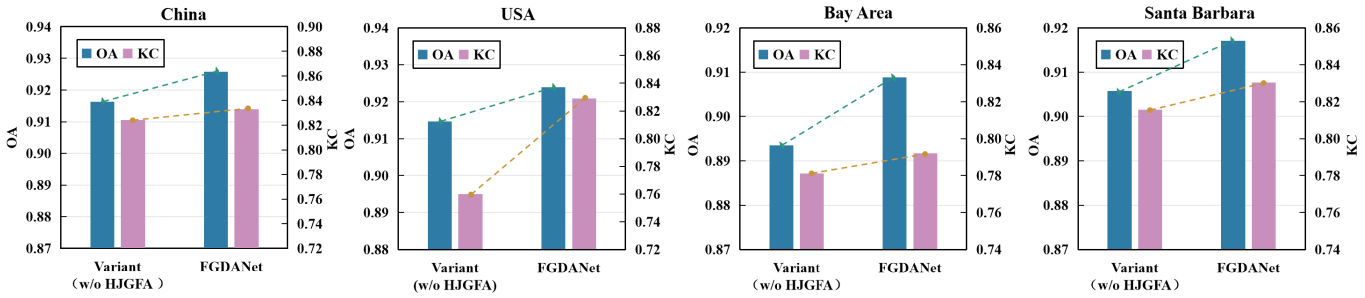


Fig. 11. CD results of FGDANet and its variant without HJGFA module on the four datasets.

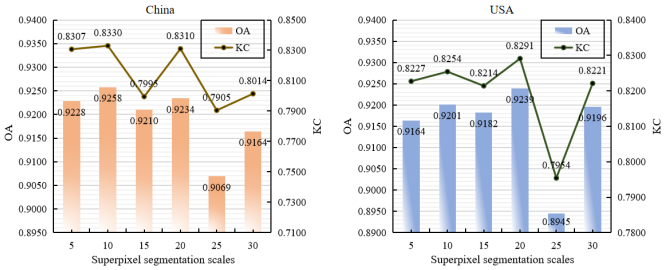


Fig. 12. CD results under different superpixel segmentation scale settings on China and USA datasets.

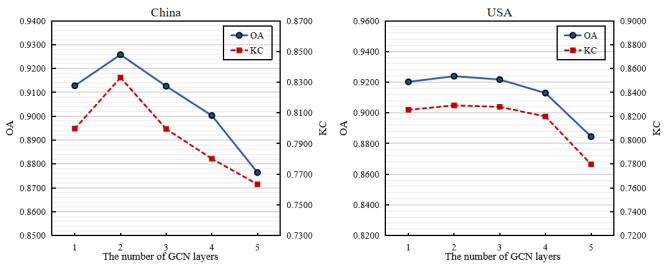


Fig. 13. CD results under different GCN layer settings on China and USA datasets.

China and USA datasets in Fig. 9. Yellow pixels represent the selected changed training samples, green pixels represent the selected unchanged training samples, and black pixels represent the discarded training samples. It is obvious that the selected changed and unchanged training samples with different proportions are all very close to the ground-truth map. This indicates that the obtained training samples have high confidence, which confirms the effectiveness of the proposed PJCS. This is because not only the consistency between the graph nodes but also the similarity between the pixels within the graph nodes is fully considered. Therefore, the reliability of the training samples can be improved.

2) Effectiveness of Graph Domain Adaptive Strategy: To verify the effectiveness of the graph domain adaptive strategy, we conducted experiments on the four widely used datasets, which were divided into two groups, i.e., the China and USA datasets, and the Bay Area and Santa Barbara datasets. We designed a variant of the proposed model without graph domain adaptive strategy [variant (w/o GDAS)] and kept other experimental settings unchanged. FGDANet and its variant were both trained with the samples from the two datasets in each group. Their parameters have been tuned to make the models achieve the best performance. Different from the variant (w/o GDAS), FGDANet consists of two domains and can obtain their CD results simultaneously. The experimental results are shown in Fig. 10. It is clear that the detection accuracy of FGDANet is higher than that of its variant in terms of OA and KC. This demonstrates that the graph domain adaptive strategy can make the model effectively utilize the information from different datasets and learn the changed and unchanged essential characteristics across domains. Thus, the feature expression capability and generalization of the model can be improved.

3) Effectiveness of Homogeneous Mutual Representation Joint Graph Feature Alignment Module: To verify the effectiveness of HJGFA module, we have conducted experiments on four datasets. Specifically, we design a variant of the proposed method without the HJGFA module, while keeping other parameters of the model unchanged. The parameters of FGDANet and its variant have been tuned to make the models achieve the best performance. The CD results are shown in Fig. 11. Compared with its variant without HJGFA module, it can be clearly seen that the proposed FGDANet can achieve better detection performance on four different datasets, which can demonstrate the effectiveness of the HJGFA module.

4) Effectiveness of Superpixel Segmentation Scale: To fully exploit the spatial–spectral correlations of pixels in HSI to construct the graph structure, in this article, we utilize SLIC to segment the original HSI into a series of superpixel regions with strong spatial–spectral correlations to determine the graph nodes of the graph structure. A smaller superpixel segmentation scale can generate more detailed segmentation results but may increase noise, while a larger superpixel segmentation scale can generate smoother segmentation but may miss some details. In order to analyze the impact of superpixel segmentation scale on detection accuracy, we conduct experiments with six different superpixel segmentation scales on China and USA datasets. As shown in Fig. 12, the detection results obtained with different superpixel segmentation scales are different. For different datasets, due to their different spatial and spectral characteristics, the corresponding optimal superpixel segmentation scales are also different. In this article, a large number of experiments have confirmed that the optimal superpixel segmentation scales of China, USA, Bay Area, and Santa Barbara datasets are 10, 20, 100, and 150, respectively.

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TABLE III
MODEL EFFICIENCY OF DIFFERENT METHODS ON CHINA DATASET

Method	CVA	PCA	IR-MAD	SVM	BCNNs	ReCNN	SGCL	HyperNet	FGDANet
Training Time (s)	-	-	-	7.94	258.73	332.59	192.27	205.80	88.55
Testing Time (s)	0.07	0.13	0.42	21.68	2.97	3.79	4.97	1.62	0.84

5) Number of Layers of GCN in Graph Convolutional Difference Feature Extractor: GCN can provide sufficient local and global features for CD by performing convolution operations on graph structures. To analyze the effect of the number of GCN layers on the detection performance, we construct the graph convolutional difference feature extractor with GCN layers from 1 to 5 on China and USA datasets. As shown in Fig. 13, the graph convolutional difference feature extractor with two GCN layers achieves the best performance, indicating that the two GCN layers have the strongest feature representation ability. This is because when the number of GCN layers is small, the model may not be able to fully extract the relationship information between the nodes in the graph, resulting in underfitting. As the number of graph convolutional layers increases, the model can extract rich features to improve the detection performance. However, when there are too many GCN layers, the features of all nodes in the same connected component may tend to be consistent, resulting in oversmoothing. Therefore, in this article, we use two GCN layers to extract the local and global features of HSI in graph convolution difference feature extractor.

6) Model Efficiency: The training time and testing time of different methods on China dataset are shown in Table III. Although the traditional methods have high detection efficiency, the detection results of these approaches are not satisfactory. On the contrary, the deep learning-based methods are more time-consuming due to the more parameters, but their performance is excellent. As shown in Table III, the proposed method can achieve not only the highest detection accuracy but also the highest computational efficiency among the competing deep learning-based methods. The reason is that the proposed method performs CD at superpixel level. Compared with the competing methods that perform CD at the pixel level, the proposed method leverages superpixel segmentation to reduce the dimensionality of the data to be processed, which avoids the pixel-by-pixel calculation. Thus, the computational complexity can be reduced while the operational efficiency can be improved.

V. CONCLUSION

In this article, a novel FGDANet for unsupervised HSI-CD is proposed. FGDANet introduces the domain adaptive ideas into unsupervised HSI-CD task, which develops a pseudosiamese framework that can not only realize effective unsupervised learning for two different domains but also achieve cross-domain information interaction to improve the generalization of the model. For each customized unsupervised learning domain, a graph convolution difference feature extractor is designed to extract the local and global difference features. Then, a PJSC is designed to fully consider the spatial–spectral correlations between pixels to obtain high-confidence training samples for unsupervised learning. To further enhance the generalization ability of FGDANet, the HJGFA module is designed to realize the cross-domain difference information interaction and improve the detection accuracy. The experimental results clearly demonstrate the effectiveness of the proposed FGDANet.

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